

21. Let the position vectors of three vertices of a triangle be $4\vec{p} + \vec{q} - 3\vec{r}$, $-5\vec{p} + \vec{q} + 2\vec{r}$ and $2\vec{p} - \vec{q} + 2\vec{r}$. If the position vectors of the orthocenter and the circumcenter of the triangle are $\frac{\vec{p} + \vec{q} + \vec{r}}{4}$ and $\alpha\vec{p} + \beta\vec{q} + \gamma\vec{r}$ respectively, then $\alpha + 2\beta + 5\gamma$ is equal to :

माना कि एक त्रिभुज के तीन शीर्षों के स्थिति सदिश $4\vec{p} + \vec{q} - 3\vec{r}$, $-5\vec{p} + \vec{q} + 2\vec{r}$ और $2\vec{p} - \vec{q} + 2\vec{r}$ हैं। यदि त्रिभुज के लम्बकेन्द्र और परिकेन्द्र के स्थिति सदिश क्रमशः $\frac{\vec{p} + \vec{q} + \vec{r}}{4}$ और $\alpha\vec{p} + \beta\vec{q} + \gamma\vec{r}$ हैं, तो $\alpha + 2\beta + 5\gamma$ बराबर है :

- (1) 6 (2) 3 (3) 1 (4) 4

Ans. B

Sol.

$$\begin{array}{ccc} & 2 & 1 \\ & \vdots & \\ \text{O} & \text{G} & \text{C} \\ \left(\frac{\vec{p} + \vec{q} + \vec{r}}{4}\right) & \left(\frac{\vec{p} + \vec{q} + \vec{r}}{3}\right) & (\alpha\vec{p} + \beta\vec{q} + \gamma\vec{r}) \end{array}$$

$$\Rightarrow \frac{2(\alpha\vec{p} + \beta\vec{q} + \gamma\vec{r}) + \left(\frac{\vec{p} + \vec{q} + \vec{r}}{4}\right)}{3} = \frac{\vec{p} + \vec{q} + \vec{r}}{3}$$

On comparing: $2\alpha + \frac{1}{4} = 1 \Rightarrow \alpha = \frac{3}{8}$

$$\beta = \gamma = \frac{3}{8}$$

$$\alpha + 2\beta + 5\gamma = 3$$

22. Let $\vec{a} = 3\hat{i} - \hat{j} + 2\hat{k}$, $\vec{b} = \vec{a} \times (\hat{i} - 2\hat{k})$ and $\vec{c} = \vec{b} \times \hat{k}$. Then the projection of $\vec{c} - 2\hat{j}$ on $\vec{a}$ is :

माना कि $\vec{a} = 3\hat{i} - \hat{j} + 2\hat{k}$, $\vec{b} = \vec{a} \times (\hat{i} - 2\hat{k})$ और $\vec{c} = \vec{b} \times \hat{k}$ है। तब $\vec{c} - 2\hat{j}$ का $\vec{a}$ पर प्रक्षेप है :

- (1) $\sqrt{14}$ (2) $2\sqrt{7}$ (3) $2\sqrt{14}$ (4) $3\sqrt{7}$

Ans. C

Sol.

$$\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & 2 \\ 1 & 0 & -2 \end{vmatrix} = 2\hat{i} + 8\hat{j} + \hat{k}$$

$$\vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 8 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 8\hat{i} - 2\hat{j}$$

Projection of $(\vec{c} - 2\hat{j})$ on $\vec{a}$

$$\Rightarrow \frac{(8\hat{i} - 4\hat{j}) \cdot (3\hat{i} - \hat{j} + 2\hat{k})}{\sqrt{9+1+4}} = 2\sqrt{14}$$

23. Let A, B, C be three points in xy-plane, whose position vector are given by $\sqrt{3}\hat{i} + \hat{j}$, $\hat{i} + \sqrt{3}\hat{j}$ and $a\hat{i} + (1-a)\hat{j}$ respectively with respect to the origin O. If the distance of the point C from the line bisecting the angle between the vectors $\vec{OA}$ and $\vec{OB}$ is $\frac{9}{\sqrt{2}}$, then the sum of all the possible values of a is :

माना कि A, B, C, xy-तल में तीन बिन्दु हैं, जिनके स्थिति सदिश, मूल-बिन्दू O के सापेक्ष, क्रमशः $\sqrt{3}\hat{i} + \hat{j}$, $\hat{i} + \sqrt{3}\hat{j}$ और $a\hat{i} + (1-a)\hat{j}$ हैं। यदि बिन्दु C की, सदिशों $\vec{OA}$ और $\vec{OB}$ के बीच के कोण को सम-द्विभाजित करने वाली रेखा से दूरी $\frac{9}{\sqrt{2}}$, तो a के सभी संभव मानों का योग है :

- (1) 0 (2) 1 (3) 2 (4) $\frac{9}{2}$

Ans. B

Sol.
$$\begin{bmatrix} \vec{A} = \sqrt{3}\hat{i} + \hat{j} \\ \vec{B} = \hat{i} + \sqrt{3}\hat{j} \\ \vec{C} = a\hat{i} + (1-a)\hat{j} \end{bmatrix}$$

Angle bisector of $\vec{OA}$ and $\vec{OB} \Rightarrow \widehat{OA} + \widehat{OB}$

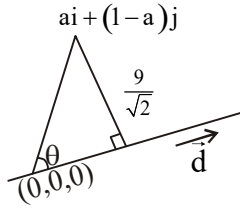
$$\Rightarrow \left(\frac{\sqrt{3}\hat{i} + \hat{j}}{2} \right) + \frac{(\hat{i} + \sqrt{3}\hat{j})}{2}$$

$$\Rightarrow \frac{(\sqrt{3}+1)\hat{i} + (\sqrt{3}+1)\hat{j}}{2}$$

$$\Rightarrow \frac{(\sqrt{3}+1)}{2}(\hat{i} + \hat{j}) \Rightarrow \text{Direction vector angle bisector}$$

$$\Rightarrow \vec{r} \Rightarrow \vec{O} + \lambda(\vec{d})$$

$$\vec{r} \Rightarrow \vec{O} + \lambda(\hat{i} + \hat{j})$$



$$\frac{\left| \left(a\hat{i} + (1-a)\hat{j} \right) \times \vec{d} \right|}{\sqrt{a^2 + (1-a)^2} |\vec{d}|} \sqrt{a^2 + (1-a)^2} \Rightarrow \left(\frac{9}{\sqrt{2}} \right)$$

$$\left| \frac{\left(a\hat{i} + (1-a)\hat{j} \right) \times (\hat{i} + \hat{j})}{\sqrt{2}} \right| = \frac{9}{\sqrt{2}}$$

$$\left| (a-1)\hat{k} + a\hat{k} \right| = 9$$

$$|2a - 1| = 9$$

$$2a - 1 = 9$$

$$a = 5$$

$$(2a - 1) = -9$$

$$2a = -8$$

$$a = -4$$

Two possible values $\Rightarrow 5 \text{ \& } -4$

$$\Rightarrow 5 + (-4) = 1$$

24. If the components of $\vec{a} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ along and perpendicular to $\vec{b} = 3\hat{i} + \hat{j} - \hat{k}$ respectively, are $\frac{16}{11}(3\hat{i} + \hat{j} - \hat{k})$ and $\frac{1}{11}(-4\hat{i} - 5\hat{j} - 17\hat{k})$, then $\alpha^2 + \beta^2 + \gamma^2$ is equal to :

यदि $\vec{a} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ के $\vec{b} = 3\hat{i} + \hat{j} - \hat{k}$ के अनुदिश और लम्बवत् घटक क्रमशः $\frac{16}{11}(3\hat{i} + \hat{j} - \hat{k})$ और $\frac{1}{11}(-4\hat{i} - 5\hat{j} - 17\hat{k})$

हैं, तो $\alpha^2 + \beta^2 + \gamma^2$ बराबर है :

(1) 16

(2) 18

(3) 26

(4) 23

Ans. C

Sol.
$$\begin{bmatrix} \vec{a} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k} \\ \vec{b} = 3\hat{i} + \hat{j} - \hat{k} \end{bmatrix}$$

$$\vec{a} = \vec{a}_b + \vec{a}_{b\perp}$$

$$\vec{a}_b \Rightarrow \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} \right) \hat{b}$$

$$\Rightarrow (\vec{a} \cdot \vec{b}) = 16$$

$$\Rightarrow (\vec{a} \cdot \vec{b}) \left(\frac{\vec{b}}{|\vec{b}|^2} \right) \Rightarrow \frac{(16)(3\hat{i} + \hat{j} - \hat{k})}{(3^2 + 1^2 + (-1)^2)}$$

$$\vec{a}_b = \frac{(\vec{a} \cdot \vec{b})}{|\vec{b}|} \left(\frac{\vec{b}}{|\vec{b}|} \right) \Rightarrow 16 \cdot \frac{(3\hat{i} + \hat{j} - \hat{k})}{11}$$

$$3\alpha + \beta - \gamma = 16$$

$$\vec{a}_{b\perp} = \vec{a} - \vec{a}_b$$

$$\Rightarrow \left(\alpha - \frac{48}{11} \right) \hat{i} + \left(\beta - \frac{16}{11} \right) \hat{j} + \left(\gamma + \frac{16}{11} \right) \hat{k}$$

$$= \left(\frac{-4}{11} \right) \hat{i} - \frac{5}{11} \hat{j} - \frac{17}{11} \hat{k}$$

$$\alpha - \frac{48}{11} = \left(-\frac{4}{11} \right)$$

$$\alpha = 4$$

$$\beta - \frac{16}{11} = \left(-\frac{5}{11} \right)$$

$$\beta = 1$$

$$\gamma + \frac{16}{11} = -\frac{17}{11}$$

$$\gamma = -3$$

25. Let $\vec{a} = 2\hat{i} - \hat{j} + 3\hat{k}$, $\vec{b} = 3\hat{i} - 5\hat{j} + \hat{k}$ and $\vec{c}$ be a vector such that $\vec{a} \times \vec{c} = \vec{c} \times \vec{b}$ and $(\vec{a} + \vec{c}) \cdot (\vec{b} + \vec{c}) = 168$. Then the maximum value of $|\vec{c}|^2$ is :

माना कि $\vec{a} = 2\hat{i} - \hat{j} + 3\hat{k}$, $\vec{b} = 3\hat{i} - 5\hat{j} + \hat{k}$ है और $\vec{c}$ एक सदिश इस प्रकार है कि $\vec{a} \times \vec{c} = \vec{c} \times \vec{b}$ और $(\vec{a} + \vec{c}) \cdot (\vec{b} + \vec{c}) = 168$ है। तब $|\vec{c}|^2$ का अधिकतम मान है :

(1) 308

(2) 154

(3) 77

(4) 462

Ans. A

Sol. $\vec{a} \times \vec{c} + \vec{b} \times \vec{c} = 0$

$$(\vec{a} + \vec{b}) \times \vec{c} = 0$$

$$\vec{c} = \lambda(\vec{a} + \vec{b})$$

$$\vec{c} = \lambda(5\hat{i} - 6\hat{j} + 4\hat{k})$$

$$\vec{c} = 5\lambda\hat{i} - 6\lambda\hat{j} + 4\lambda\hat{k}$$

$$\left((2+5\lambda)\hat{i} - (1+6\lambda)\hat{j} + (4\lambda+3)\hat{k} \right) \cdot \left((3+5\lambda)\hat{i} - (6\lambda+5)\hat{j} + (1+4\lambda)\hat{k} \right) = 168$$

$$(2+5\lambda)(3+5\lambda) + (1+6\lambda)(6\lambda+5) + (4\lambda+3)(1+4\lambda) = 168$$

$$\lambda^2 + \lambda - 2 = 0$$

$$(\lambda + 2)(\lambda - 1) = 0$$

$$\lambda = -2 \text{ or } \lambda = 1$$

$$|\vec{c}| = \sqrt{100 + 144 + 64}$$

$$|\vec{c}|^2 = 244 + 64 = 308$$

26. Let $L_1 : \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $L_2 : \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$ be two lines. Then which of the following points lies on the line of the shortest distance between L_1 and L_2 :

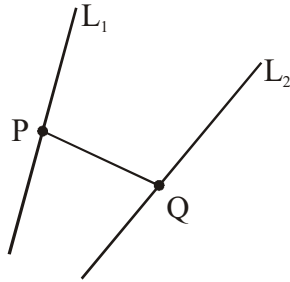
माना कि $L_1 : \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ और $L_2 : \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$ दो रेखाएँ हैं। तब निम्नलिखित में से कौनसा बिन्दु L_1

और L_2 के बीच लघुतम दूरी की रेखा पर स्थित है ?

- (1) $\left(-\frac{5}{3}, -7, 1\right)$ (2) $\left(\frac{8}{3}, -1, \frac{1}{3}\right)$ (3) $\left(2, 3, \frac{1}{3}\right)$ (4) $\left(\frac{14}{3}, -3, \frac{22}{3}\right)$

Ans. D

Sol.



$$P(2\lambda + 1, 3\lambda + 2, 4\lambda + 3) \text{ on } L_1$$

$$Q(3\mu + 2, 4\mu + 4, 5\mu + 5) \text{ on } L_2$$

$$\text{Dr's of } PQ = 3\mu - 2\lambda + 1, 4\mu - 3\lambda + 2, 5\mu - 4\lambda + 2$$

$$PQ \perp L_1$$

$$\Rightarrow (3\mu - 2\lambda + 1)2 + (4\mu - 3\lambda + 2)3 + (5\mu - 4\lambda + 2)4 = 0$$

$$38\mu - 29\lambda + 16 = 0 \quad \dots(1)$$

$$PQ \perp L_2$$

$$\Rightarrow (3\mu - 2\lambda + 1)3 + (4\mu - 3\lambda + 2)4 + (5\mu - 4\lambda + 2)5 = 0$$

$$50\mu - 38\lambda + 21 = 0 \quad \dots(2)$$

By (1) and (2)

$$\lambda = \frac{1}{3}; \mu = \frac{-1}{6}$$

$$\therefore P\left(\frac{5}{3}, 3, \frac{13}{3}\right) \& Q\left(\frac{3}{2}, \frac{10}{3}, \frac{25}{6}\right)$$

Line PQ

$$\frac{x - \frac{5}{3}}{\frac{1}{6}} = \frac{y - 3}{\frac{-1}{3}} = \frac{z - \frac{13}{3}}{\frac{1}{6}}$$

$$\frac{x - \frac{5}{3}}{1} = \frac{y - 3}{-2} = \frac{z - \frac{13}{3}}{1}$$

$$\text{Point} \left(\frac{14}{3}, -3, \frac{22}{3} \right)$$

lies on the line PQ.

27. Let in a ΔABC , the length of the side AC be 6, the vertex B be (1, 2, 3) and the vertices A, C lie on the line

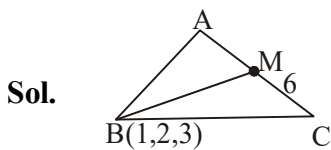
$$\frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2}. \text{ Then the area (in sq. units) of } \Delta ABC \text{ is :}$$

माना कि एक ΔABC में भुजा AC की लम्बाई 6 है, शीर्ष B(1, 2, 3) है और शीर्ष A, C रेखा $\frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2}$ पर स्थित हैं।

तब ΔABC का क्षेत्रफल (वर्ग इकाई में) है :

- (1) 17 (2) 21 (3) 56 (4) 42

Ans. B



$$\text{Let } M(3\lambda + 6, 2\lambda + 7, -2\lambda + 7)$$

$$\overrightarrow{BM} = (3\lambda + 5)\hat{i} + (2\lambda + 5)\hat{j} + (-2\lambda + 4)\hat{k}$$

$$\overrightarrow{AC} \cdot \overrightarrow{BM} = 0 = 3(3\lambda + 5) + 2(2\lambda + 5) - 2(-2\lambda + 4)$$

$$\overrightarrow{BM} = 2\hat{i} + 3\hat{j} + 6\hat{k}$$

$$|\overrightarrow{BM}| = 7$$

$$\text{Area} = \frac{1}{2} \times 6 \times 7 = 21$$

28. Let the line passing through the points $(-1, 2, 1)$ and parallel to the line $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z}{4}$ intersect the line

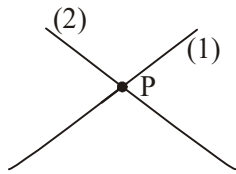
$\frac{x+2}{3} = \frac{y-3}{2} = \frac{z-4}{1}$ at the point P. Then the distance of P from the point $Q(4, -5, 1)$ is :

माना कि बिन्दु $(-1, 2, 1)$ से होकर गुजरती और रेखा $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z}{4}$ के समान्तर रेखा, रेखा $\frac{x+2}{3} = \frac{y-3}{2} = \frac{z-4}{1}$ को बिन्दु P पर प्रतिच्छेद करती है। तब बिन्दु $Q(4, -5, 1)$ से P की दूरी है :

- (1) $5\sqrt{5}$ (2) 10 (3) 5 (4) $5\sqrt{6}$

Ans. A

Sol.



Equation of line through point $(-1, 2, 1)$ is $\rightarrow$

$$\Rightarrow \frac{x+1}{2} = \frac{y-2}{3} = \frac{z-1}{4} = \lambda$$

$$\text{So, } \begin{cases} x = 2\lambda - 1 \\ y = 3\lambda + 2 \\ z = 4\lambda + 1 \end{cases}$$

$$\text{By (1)} \rightarrow \frac{x+2}{3} = \frac{y-3}{2} = \frac{z-4}{1} = \mu \text{ (Let)}$$

$$\text{So, } \begin{cases} x = 3\mu - 2 \\ y = 2\mu + 3 \\ z = \mu + 4 \end{cases}$$

For intersection point 'P'

$$x = 2\lambda - 1 = 3\mu - 2$$

$$y = 3\lambda + 2 = 2\mu + 3 \quad \begin{cases} \lambda = 1 \\ \mu = 1 \end{cases}$$

$$z = 4\lambda + 1 = \mu + 4$$

So, point $P(x, y, z) = (1, 5, 5)$

W& $Q(4, -5, 1)$

$$\begin{aligned} \therefore PQ &= \sqrt{9+100+16} \\ &= \sqrt{125} = 5\sqrt{5} \end{aligned}$$

29. Let $A(x, y, z)$ be a point in xy -plane, which is equidistant from three points $(0, 3, 2)$, $(2, 0, 3)$ and $(0, 0, 1)$.

Let $B = (1, 4, -1)$ and $C = (2, 0, -2)$. Then among the statements

(S1) : ΔABC is an isosceles right angled triangle, and

(S2) : the area of ΔABC is $\frac{9\sqrt{2}}{2}$:

(1) both are true (2) both are false (3) only (S2) is true (4) only (S1) is true

माना कि xy -तल में एक बिन्दु $A(x, y, z)$ है जो कि तीन बिन्दुओं $(0, 3, 2)$, $(2, 0, 3)$ और $(0, 0, 1)$ से समदूरस्थ है।

माना कि $B = (1, 4, -1)$ और $C = (2, 0, -2)$ है। तब कथनों

(S1) : ΔABC एक समद्विबाहु समकोण त्रिभुज है, और

(S2) : ΔABC का क्षेत्रफल $\frac{9\sqrt{2}}{2}$ है

में से

(1) दोनों सत्य हैं (2) दोनों असत्य हैं (3) केवल (S2) सत्य है (4) केवल (S1) सत्य है

Ans. D

Sol. $A(x, y, z)$ Let $P(0,3,2)$, $Q(2,0,3)$, $R(0,0,1)$

$$AP = AQ = AR$$

$$x^2 + (y - 3)^2 + (z - 2)^2 = (x - 2)^2 + y^2 + (z - 3)^2 = x^2 + y^2 + (z - 1)^2$$

In xy -plane $z = 0$

$$\text{So, } x^2 - 4x + 4 + y^2 + 9 = x^2 + y^2 + 1$$

$$x = 3$$

$$9 + y^2 - 6y + 9 + 4 = x^2 + y^2 + 1$$

So, $A(3,2,0)$ also $B(1,4,-1)$ & $C(2,0,-2)$

$$\text{Now, } AB = \sqrt{4 + 4 + 1} = 3$$

$$AC = \sqrt{1 + 4 + 4} = 3$$

$$BC = \sqrt{1 + 16 + 1} = \sqrt{18}$$

$$AB = AC$$

$$\text{isosceles } \Delta \text{ \& } AB^2 + AC^2 = BC^2$$

$$\text{Area of } \Delta ABC = \frac{1}{2} \times \text{base} \times \text{height}$$

$$\frac{1}{2} \times 3 \times 3 = \frac{9}{2}$$

So, only S_1 is true

30. Let $L_1 : \frac{x-1}{1} = \frac{y-2}{-1} = \frac{z-1}{2}$ and $L_2 : \frac{x+1}{-1} = \frac{y-2}{2} = \frac{z}{1}$ be two lines. Let L_3 be a line passing through the point (α, β, γ) and be perpendicular to both L_1 and L_2 . If L_3 intersects L_1 , then $|5\alpha - 11\beta - 8\gamma|$ is equals :

माना कि $L_1 : \frac{x-1}{1} = \frac{y-2}{-1} = \frac{z-1}{2}$ और $L_2 : \frac{x+1}{-1} = \frac{y-2}{2} = \frac{z}{1}$ दो रेखाएँ हैं। माना L_3 , बिन्दु (α, β, γ) से गुजरने वाली और L_1 तथा L_2 दोनों के लम्बवत् एक रेखा है। यदि L_3 , L_1 को प्रतिच्छेद करती है, तो $|5\alpha - 11\beta - 8\gamma|$ बराबर है :

- (1) 16 (2) 25 (3) 20 (4) 18

Ans. B

Sol. $L_1 : \frac{x-1}{1} = \frac{y-2}{-1} = \frac{z-1}{2}$

$$L_1 : \vec{r} = (1, 2, 1) + \lambda(1, -1, 2)$$

$$L_2 : \frac{x+1}{-1} = \frac{y-2}{2} = \frac{z}{1}$$

$$L_2 : \vec{r} = (-1, 2, 0) + \mu(-1, 2, 1)$$

$$\vec{P} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 2 \\ -1 & 2 & 1 \end{vmatrix}$$

$$= \hat{i}(-5) - \hat{j}(3) + \hat{k}(1)$$

$$= -5\hat{i} - 3\hat{j} + \hat{k}$$

$$P(\alpha, \beta, \gamma) \quad Q(1 + \lambda, 2 - \lambda, 1 + 2\lambda)$$

$$\overrightarrow{PQ} = (1 + \lambda - \alpha, 2 - \lambda - \beta, 1 + 2\lambda - \gamma)$$

$$1 + \lambda - \alpha = -5\mu \quad 2 - \lambda - \beta = -3\mu \quad 1 + 2\lambda - \gamma = \mu$$

$$1 + \lambda + 5\mu = \alpha \quad 2 - \lambda + 3\mu = \beta \quad \gamma = 1 + 2\lambda - \mu$$

$$|5\alpha - 11\beta - 8\gamma| = 25$$

31. Let P be the foot of the perpendicular from point $(1, 2, 2)$ on the line $L : \frac{x-1}{1} = \frac{y+1}{-1} = \frac{z-2}{2}$. Let the line $\vec{r} = (-\hat{i} + \hat{j} - 2\hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k}), \lambda \in \mathbb{R}$, intersect the line L, at Q. Then $2(PQ)^2$ is equal to :

माना कि P, बिन्दु $(1, 2, 2)$ से रेखा $L : \frac{x-1}{1} = \frac{y+1}{-1} = \frac{z-2}{2}$ पर लम्ब का पाद है। माना कि रेखा $\vec{r} = (-\hat{i} + \hat{j} - 2\hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k}), \lambda \in \mathbb{R}$, रेखा L को Q पर काटती है। तब $2(PQ)^2$ बराबर है :

$$\vec{r} = (-\hat{i} + \hat{j} - 2\hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k}), \lambda \in \mathbb{R}, \text{ रेखा } L \text{ को } Q \text{ पर काटती है। तब } 2(PQ)^2 \text{ बराबर है :}$$

(1) 19

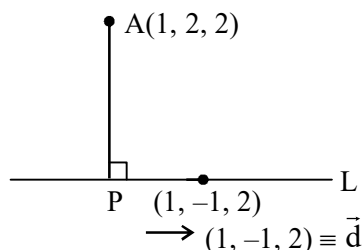
(2) 27

(3) 25

(4) 29

Ans. B

Sol.



$$L: \frac{x-1}{1} = \frac{y+1}{-1} = \frac{z-2}{2} = \mu$$

$$= \mu P(\mu+1, -\mu-1, 2\mu+2)$$

$$\overrightarrow{AP} \cdot \vec{d} = 0 \Rightarrow (\mu, -\mu-3, 2\mu) \cdot (1, -1, 2) = 0$$

$$\Rightarrow \mu + \mu + 3 + 4\mu = 0 \Rightarrow \mu = -\frac{1}{2}$$

$$\therefore P\left(\frac{-1}{2}+1, +\frac{1}{2}-1, 2\left(\frac{-1}{2}\right)+2\right)$$

$$P\left(\frac{1}{2}, \frac{-1}{2}, 1\right)$$

Now general pt. on L_2 is $Q(-1+\lambda, 1-\lambda, -2+\lambda)$

Equate it with general pt of L

$$\begin{array}{l|l|l} \mu+1=-1+\lambda & -\mu-1=1-\lambda & 2\mu+2=-2+\lambda \\ \mu=\lambda-2 & \mu=\lambda-2 & \downarrow \\ & & 2(\lambda-2)+2=-2+\lambda \\ & & 2\lambda-4+2=-2+\lambda \end{array}$$

$$\therefore \mu = -2, \lambda = 0$$

$$\therefore Q \equiv -2, \lambda = 0$$

$$P\left(\frac{1}{2}, \frac{-1}{2}, 1\right) \text{ and } Q(-1, 1, -2)$$

$$PQ = \sqrt{\left(\frac{1}{2}+1\right)^2 + \left(\frac{-1}{2}-1\right)^2 + (1+2)^2}$$

$$= \sqrt{\frac{9}{4} + \frac{9}{4} + 9} = \sqrt{\frac{54}{4}}$$

$$\therefore 2(PQ)^2 = 2\left(\frac{54}{4}\right) = 27$$

32. Let a unit vector which makes an angle of 60° with $2\hat{i} + 2\hat{j} - \hat{k}$ and an angle of 45° with $\hat{i} - \hat{k}$ be $\vec{C}$. Then

$$\vec{C} + \left(-\frac{1}{2}\hat{i} + \frac{1}{3\sqrt{2}}\hat{j} - \frac{\sqrt{2}}{3}\hat{k} \right) \text{ is :}$$

माना सदिश $2\hat{i} + 2\hat{j} - \hat{k}$ से 60° का कोण तथा सदिश $\hat{i} - \hat{k}$ से 45° का कोण बनाने वाला एक मात्रक सदिश $\vec{C}$ है। तो

$$\vec{C} + \left(-\frac{1}{2}\hat{i} + \frac{1}{3\sqrt{2}}\hat{j} - \frac{\sqrt{2}}{3}\hat{k} \right) \text{ है :}$$

$$(1) \left(\frac{1}{\sqrt{3}} + \frac{1}{2} \right) \hat{i} + \left(\frac{1}{\sqrt{3}} - \frac{1}{3\sqrt{2}} \right) \hat{j} + \left(\frac{1}{\sqrt{3}} + \frac{\sqrt{2}}{3} \right) \hat{k}$$

$$(2) \frac{\sqrt{2}}{3} \hat{i} + \frac{1}{3\sqrt{2}} \hat{j} - \frac{1}{2} \hat{k}$$

$$(3) \frac{\sqrt{2}}{3} \hat{i} - \frac{1}{2} \hat{k}$$

$$(4) -\frac{\sqrt{2}}{3} \hat{i} + \frac{\sqrt{2}}{3} \hat{j} + \left(\frac{1}{2} + \frac{2\sqrt{2}}{3} \right) \hat{k}$$

Ans. C

Sol. Let $\vec{C} = a\hat{i} + b\hat{j} + c\hat{k}$

$$(a\hat{i} + b\hat{j} + c\hat{k}) \cdot (2\hat{i} + 2\hat{j} - \hat{k}) = 1 \times 3 \times \frac{1}{2}$$

$$2a + 2b - c = \frac{3}{2} \quad \text{_____}(1)$$

$$(a\hat{i} + b\hat{j} + c\hat{k}) \cdot (\hat{i} - \hat{k}) = 1 \times \sqrt{2} \times \frac{1}{\sqrt{2}}$$

$$a - c = 1 \quad \text{_____}(2)$$

$$a^2 + b^2 + c^2 = 1 \quad \text{_____}(3)$$

Solving (1), (2) and (3)

$$a + 2b = \frac{1}{2}$$

$$a^2 + b^2 + (a - 1)^2 = 1$$

$$2a^2 - 2a + b^2 = 0$$

$$2a^2 - 2a + \left(\frac{2a - 1}{4} \right)^2 = 0$$

$$32a^2 - 32a + 4a^2 - 4a + 1 = 0$$

$$36a^2 - 36a + 1 = 0$$

$$a = \frac{36 \pm \sqrt{(36)^2 - 4(36)}}{2 \times 36}$$

$$= \frac{1}{2} \pm \frac{\sqrt{2}}{3}$$

$$b = \frac{1-2a}{4} \Rightarrow b = \frac{1 \pm \frac{2\sqrt{2}}{3} - 1}{4}$$

$$= -\pm \frac{1}{3\sqrt{2}}$$

$$C = -\frac{1}{2} \pm \frac{\sqrt{2}}{3}$$

$$C + \left(\frac{-1}{2} \hat{i} + \frac{1}{3\sqrt{2}} \hat{j} - \frac{\sqrt{2}}{3} \hat{k} \right)$$

$$= \frac{\sqrt{2}}{3} \hat{i} - \frac{1}{2} \hat{k}$$

33. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\vec{c} = x\hat{i} + 2\hat{j} + 3\hat{k}$, $x \in \mathbb{R}$. If $\vec{d}$ is the unit vector in the direction of $\vec{b} + \vec{c}$ such that $\vec{a} \cdot \vec{d} = 1$, then $(\vec{a} \times \vec{b}) \cdot \vec{c}$ is equal to :

माना $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ तथा $\vec{c} = x\hat{i} + 2\hat{j} + 3\hat{k}$, $x \in \mathbb{R}$ है। यदि $\vec{b} + \vec{c}$ की दिशा में एक इकाई सदिश $\vec{d}$ इस प्रकार है कि $\vec{a} \cdot \vec{d} = 1$, तो $(\vec{a} \times \vec{b}) \cdot \vec{c}$ बराबर है :

(1) 11

(2) 3

(3) 6

(4) 9

Ans. A

Sol. $\vec{d} = \lambda((2+x)\hat{i} + 6\hat{j} - 2\hat{k})$

$$\vec{a} \cdot \vec{d} = 1 \Rightarrow \lambda(x+6) = 1 \quad \text{_____ (1)}$$

$$\text{and } |\vec{d}| = 1 \Rightarrow \lambda^2((x+2)^2 + 36 + 4) = 1 \quad \text{_____ (2)}$$

(1) and (2) : $x = 1$

$$\vec{c} = \hat{i} + 2\hat{j} + 3\hat{k}$$

$$\vec{a} \times \vec{b} = -9\hat{i} + 7\hat{j} + 2\hat{k}$$

$$(\vec{a} \times \vec{b}) \cdot \vec{c} = -9 + 14 + 6 = 11$$

34. Consider three vectors $\vec{a}, \vec{b}, \vec{c}$. Let $|\vec{a}| = 2, |\vec{b}| = 3$ and $\vec{a} = \vec{b} \times \vec{c}$. If $\alpha \in \left[0, \frac{\pi}{3}\right]$ is the angle between the vectors $\vec{b}$ and $\vec{c}$, then the minimum value of $27|\vec{c} - \vec{a}|^2$ is equal to :

तीन सदिशों $\vec{a}, \vec{b}, \vec{c}$ का विचार कीजिए। माना $|\vec{a}| = 2, |\vec{b}| = 3$ तथा $\vec{a} = \vec{b} \times \vec{c}$ हैं। यदि सदिशों $\vec{b}$ तथा $\vec{c}$ के बीच का कोण $\alpha \in \left[0, \frac{\pi}{3}\right]$ है, तो $27|\vec{c} - \vec{a}|^2$ का न्यूनतम मान है :

- (1) 105 (2) 124 (3) 121 (4) 110

Ans. B

Sol. $\vec{a} = \vec{b} \times \vec{c}$

$$|\vec{a}| = 2, |\vec{b}| = 3$$

$$\vec{a} \cdot \vec{b} = 0 \text{ and } \vec{a} \cdot \vec{c} = 0$$

$$|\vec{c} - \vec{a}|^2 = |\vec{c}|^2 + |\vec{a}|^2 - 2\vec{c} \cdot \vec{a}$$

$$= 4 + |\vec{c}|^2$$

$$|\vec{a}| = |\vec{b} \times \vec{c}| = |\vec{b}| \sin \alpha |\vec{c}|$$

$$\Rightarrow \sin \alpha |\vec{c}| = \frac{2}{3}$$

$$\Rightarrow \sin^2 \alpha = \frac{4}{9|\vec{c}|^2}$$

$$\Rightarrow |\vec{c}|^2 = \frac{4}{9\sin^2 \alpha}$$

$$\Rightarrow |\vec{c} - \vec{a}|^2 = 4 + \frac{4}{9\sin^2 \alpha}$$

$$\text{For } |\vec{c} - \vec{a}|^2 \text{ to be minimum for } \alpha \in \left[0, \frac{\pi}{3}\right]$$

$$\sin \alpha = \frac{\sqrt{3}}{2}$$

$$27|\vec{c} - \vec{a}|^2 = 27 \left[4 + \frac{4.4}{9.3} \right]$$

$$= 27 \left[\frac{124}{27} \right] = 124$$

35. Let $\vec{a} = 6\hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} + \hat{j}$. If $\vec{c}$ is a vector such that $|\vec{c}| \geq 6$, $\vec{a} \cdot \vec{c} = 6|\vec{c}|$, $|\vec{c} - \vec{a}| = 2\sqrt{2}$ and the angle between $\vec{a} \times \vec{b}$ and $\vec{c}$ is 60° , then $|(\vec{a} \times \vec{b}) \times \vec{c}|$ is equal to :

माना $\vec{a} = 6\hat{i} + \hat{j} - \hat{k}$ तथा $\vec{b} = \hat{i} + \hat{j}$ है। यदि एक सदिश $\vec{c}$ इस प्रकार है कि $|\vec{c}| \geq 6$, $\vec{a} \cdot \vec{c} = 6|\vec{c}|$, $|\vec{c} - \vec{a}| = 2\sqrt{2}$ तथा $\vec{a} \times \vec{b}$ और $\vec{c}$ के मध्य का कोण 60° है, तो $|(\vec{a} \times \vec{b}) \times \vec{c}|$ बराबर है :

- (1) $\frac{9}{2}(6 - \sqrt{6})$ (2) $\frac{9}{2}(6 + \sqrt{6})$ (3) $\frac{3}{2}\sqrt{3}$ (4) $\frac{3}{2}\sqrt{6}$

Ans. B

Sol. $|(\vec{a} \times \vec{b}) \times \vec{c}| = |\vec{a} \times \vec{b}| |\vec{c}| \frac{\sqrt{3}}{2}$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 6 & 1 & -1 \\ 1 & 1 & 0 \end{vmatrix} = \hat{i} - \hat{j} + 5\hat{k}$$

$$|\vec{a} \times \vec{b}| = \sqrt{27}$$

$$\therefore |\vec{c} - \vec{a}| = 2\sqrt{2}$$

$$|\vec{c}|^2 + |\vec{a}|^2 - 2\vec{c} \cdot \vec{a} = 8$$

$$|\vec{c}|^2 + |\vec{a}|^2 - 12|\vec{c}| = 8$$

$$|\vec{c}|^2 - 12|\vec{c}| + 30 = 0$$

$$|\vec{c}| = 6 + \sqrt{6}$$

$$\therefore |\vec{c}| > 6$$

$$|(\vec{a} \times \vec{b}) \times \vec{c}| = \sqrt{27} \times (6 + \sqrt{6}) \times \frac{\sqrt{3}}{2} = \frac{9}{2}(6 + \sqrt{6})$$

36. Let $P(x, y, z)$ be a point in the first octant, whose projection in the xy -plane is the point Q . Let $OP = \gamma$; the angle between OQ and the positive x -axis is θ ; and the angle between OP and the positive z -axis be ϕ , where O is the origin. Then the distance of P from the x -axis is :

माना $P(x, y, z)$ प्रथम अष्टांशक में एक बिन्दु है जिसका xy -तल में प्रक्षेप बिन्दु Q है। माना $OP = \gamma$ है तथा OQ और धनात्मक

x-axis कोण θ है एवं OP तथा घनात्मक z-अक्ष के बीच कोण ϕ है, जहाँ O मूल बिन्दु है, तब P की x-अक्ष से दूरी होगी :

(1) $\gamma\sqrt{1+\cos^2\theta\sin^2\phi}$

(2) $\gamma\sqrt{1+\cos^2\phi\sin^2\theta}$

(3) $\gamma\sqrt{1-\sin^2\phi\cos^2\theta}$

(4) $\gamma\sqrt{1-\sin^2\theta\cos^2\phi}$

Ans. C

Sol. $\vec{OP} = x\hat{i} + y\hat{j} + z\hat{k}$

$\vec{OQ} = x\hat{i} + y\hat{j}$

$|\vec{OP}| = \gamma = \sqrt{x^2 + y^2 + z^2}$

$\cos\theta = \frac{x}{\sqrt{x^2 + y^2}} \Rightarrow \cos^2\theta = \frac{x^2}{\gamma^2 - z^2} = \frac{x^2}{\gamma^2 - \gamma^2\cos^2\phi}$

$\cos\phi = \frac{z}{\sqrt{x^2 + y^2 + z^2}} = \frac{z}{\gamma}$

Distance of P from x-axis = $\sqrt{y^2 + z^2}$

$d = \sqrt{\gamma^2 - x^2}$

$\Rightarrow x^2 = \gamma^2 \sin^2\phi \cos^2\theta$

$\Rightarrow d = \sqrt{\gamma^2 - \gamma^2 \sin^2\phi \cos^2\theta}$

$= \gamma\sqrt{1 - \sin^2\phi \cos^2\theta}$

37. Let $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = 2\hat{i} + 3\hat{j} - 5\hat{k}$ and $\vec{c} = 3\hat{i} - \hat{j} + \lambda\hat{k}$ be three vectors. Let $\vec{r}$ be a unit vector along $\vec{b} + \vec{c}$. If $\vec{r} \cdot \vec{a} = 3$, then 3λ is equal to :

माना तीन सदिश $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = 2\hat{i} + 3\hat{j} - 5\hat{k}$ तथा $\vec{c} = 3\hat{i} - \hat{j} + \lambda\hat{k}$ हैं। माना $\vec{b} + \vec{c}$ के अनुदिश एक मात्रक सदिश $\vec{r}$ है।

यदि $\vec{r} \cdot \vec{a} = 3$ है, तो 3λ बराबर है :

(1) 30

(2) 21

(3) 25

(4) 27

Ans. C

Sol. $\vec{r} = k(\vec{b} + \vec{c})$

$\vec{r} \cdot \vec{a} = 3$

$\vec{r} \cdot \vec{a} = k(\vec{b} \cdot \vec{a} + \vec{c} \cdot \vec{a})$

$$3 = k(2 + 6 - 15 + 3 - 2 + 3\lambda)$$

$$3 = k(-3 + 3\lambda) \quad \dots(1)$$

$$\vec{r} = k(5\hat{i} - 2\hat{j} - (5 - \lambda)\hat{k})$$

$$|\vec{r}| = k\sqrt{25 + 4 + 25 + \lambda^2 - 10\lambda}$$

$$k = \frac{3}{-6 + 3\lambda} = \frac{1}{-2 + \lambda} \text{ put in (1)}$$

$$4 + \lambda^2 - 4\lambda = (54 + \lambda^2 - 10\lambda)$$

$$6\lambda = 50$$

$$3\lambda = 25$$

38. Let three vectors $\vec{a} = \alpha\hat{i} + 4\hat{j} + 2\hat{k}$, $\vec{b} = 5\hat{i} + 3\hat{j} + 4\hat{k}$, $\vec{c} = x\hat{i} + y\hat{j} + z\hat{k}$ form a triangle such that $\vec{c} = \vec{a} - \vec{b}$ and the area of the triangle is $5\sqrt{6}$. If α is a positive real number, then $|\vec{c}|^2$ is equal to :

माना तीन सदिश $\vec{a} = \alpha\hat{i} + 4\hat{j} + 2\hat{k}$, $\vec{b} = 5\hat{i} + 3\hat{j} + 4\hat{k}$, $\vec{c} = x\hat{i} + y\hat{j} + z\hat{k}$ एक त्रिभुज बनाते हैं, जो इस प्रकार है कि $\vec{c} = \vec{a} - \vec{b}$ है तथा इस त्रिभुज का क्षेत्रफल $5\sqrt{6}$ है। यदि α एक धनात्मक वास्तविक संख्या है, तो $|\vec{c}|^2$ बराबर है :

- (1) 14 (2) 10 (3) 12 (4) 16

Ans. A

Sol. $\vec{c} = (\alpha - 5)\hat{i} + \hat{j} - 2\hat{k}$

$$\text{Area of } \Delta ABC = \left| \frac{1}{2} (\vec{a} \times \vec{b}) \right| = 5\sqrt{6} \Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \alpha & 4 & 2 \\ 5 & 3 & 4 \end{vmatrix} = 10\sqrt{6}$$

$$|10\hat{i} - (4\alpha - 10)\hat{j} + (3\alpha - 20)\hat{k}| = 10\sqrt{6}$$

$$\Rightarrow \sqrt{100 + (10 - 4\alpha)^2 + (3\alpha - 20)^2} = 10\sqrt{6}$$

$$\Rightarrow 25\alpha^2 - 200\alpha + 600 = 600$$

$$\Rightarrow \alpha = 8 \quad (\because \alpha \in \mathbb{R}^+)$$

$$\vec{c} = 3\hat{i} + \hat{j} - 2\hat{k}$$

$$\Rightarrow |\vec{c}|^2 = 9 + 1 + 4 = 14$$

39. Between the following two statements :

Statement I : Let $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} - \hat{k}$. Then the vector $\vec{r}$ satisfying $\vec{a} \times \vec{r} = \vec{a} \times \vec{b}$ and $\vec{a} \cdot \vec{r} = 0$ is of magnitude $\sqrt{10}$.

Statement II : In a triangle ABC, $\cos 2A + \cos 2B + \cos 2C \geq -\frac{3}{2}$.

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

निम्नलिखित दो कथनों :

कथन I : माना $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$ और $\vec{b} = 2\hat{i} + \hat{j} - \hat{k}$ हैं। तो $\vec{a} \times \vec{r} = \vec{a} \times \vec{b}$ तथा $\vec{a} \cdot \vec{r} = 0$ को संतुष्ट करने वाले सदिश $\vec{r}$ का परिमाण $\sqrt{10}$ है।

कथन II : एक त्रिभुज ABC में $\cos 2A + \cos 2B + \cos 2C \geq -\frac{3}{2}$ है।

- (1) कथन I सही है परन्तु कथन II गलत है।
- (2) कथन I गलत है परन्तु II सही है।
- (3) कथन I तथा कथन II दोनों सही है।
- (4) कथन I तथा कथन II गलत है।

Ans. B

Sol. $\forall$ two vectors $\vec{c}$ and $\vec{d}$

$$|\vec{c} \times \vec{d}|^2 = |\vec{c}|^2 |\vec{d}|^2 - (\vec{c} \cdot \vec{d})^2$$

replacing $\vec{c} = \vec{a}$ and $\vec{d} = \vec{r}$

$$\Rightarrow |\vec{a} \times \vec{r}|^2 = |\vec{a}|^2 |\vec{r}|^2 - (\vec{a} \cdot \vec{r})^2$$

$$\Rightarrow |\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{r}|^2$$

$$(\because \vec{a} \times \vec{r} = \vec{a} \times \vec{b} \text{ and } \vec{a} \cdot \vec{r} = 0)$$

$$\Rightarrow 35 = 14 |\vec{r}|^2$$

$$\Rightarrow |\vec{r}| = \sqrt{\frac{35}{14}} = \sqrt{\frac{5}{2}} \neq \sqrt{10}$$

∴ Statement {I} is incorrect Statement II is correct

$$\left(\text{i.e., } \cos 2A + \cos 2B + \cos 2C \geq -\frac{3}{2} \right)$$

Proof: ∴ $(\vec{OA} + \vec{OB} + \vec{OC}) \geq 0$ _____ (1)

and $|\vec{OA}|^2 = |\vec{OB}|^2 = |\vec{OC}|^2 = R^2$ _____ (2)

Now, using (1), we get $|\vec{OA}|^2 + |\vec{OB}|^2 + |\vec{OC}|^2 + 2(\vec{OA} \cdot \vec{OB} + \vec{OB} \cdot \vec{OC} + \vec{OC} \cdot \vec{OA}) \geq 0$

$$\Rightarrow 3R^2 + 2R^2(\cos 2A + \cos 2B + \cos 2C) \geq 0$$

$$\Rightarrow \cos 2A + \cos 2B + \cos 2C \geq -\frac{3}{2}$$

40. Let $\vec{a} = 2\hat{i} + \alpha\hat{j} + \hat{k}$, $\vec{b} = -\hat{i} + \hat{k}$, $\vec{c} = \beta\hat{j} - \hat{k}$, where α and β are integers and $\alpha\beta = -6$. Let the values of the ordered pair (α, β) , for which the area of the parallelogram of diagonals $\vec{a} + \vec{b}$ and $\vec{b} + \vec{c}$ is $\frac{\sqrt{21}}{2}$, be (α_1, β_1) and (α_2, β_2) . Then $\alpha_1^2 + \beta_1^2 - \alpha_2\beta_2$ is equal to:

माना $\vec{a} = 2\hat{i} + \alpha\hat{j} + \hat{k}$, $\vec{b} = -\hat{i} + \hat{k}$, $\vec{c} = \beta\hat{j} - \hat{k}$ हैं, जहाँ α और β पूर्णांक हैं और $\alpha\beta = -6$ है। माना क्रमित युग्म (α, β) के मान जिनके लिए विकर्ण $\vec{a} + \vec{b}$ और $\vec{b} + \vec{c}$ के समांतर चतुर्भुज का क्षेत्रफल $\frac{\sqrt{21}}{2}$ है, (α_1, β_1) और (α_2, β_2) हैं। तो $\alpha_1^2 + \beta_1^2 - \alpha_2\beta_2$ बराबर है :

(1) 21

(2) 17

(3) 19

(4) 24

Ans. C

Sol. Area of parallelogram $= \frac{1}{2} |\vec{d}_1 \times \vec{d}_2|$

$$A = \frac{1}{2} |(\vec{a} + \vec{b}) \times (\vec{b} + \vec{c})|$$

$$= \frac{\sqrt{21}}{2}$$

so, $\vec{a} + \vec{b} = \hat{i} + \alpha\hat{j} + 2\hat{k}$

$$\vec{b} + \vec{c} = -\hat{i} + \beta\hat{j}$$

$$(\vec{a} + \vec{b}) \times (\vec{b} + \vec{c}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & \alpha & 2 \\ -1 & \beta & 0 \end{vmatrix}$$

$$= \hat{i}(-2\beta) - \hat{j}(2) + \hat{k}(\beta + \alpha)$$

$$|(\vec{a} + \vec{b}) \times (\vec{b} + \vec{c})| = \sqrt{4\beta^2 + 4 + (\alpha + \beta)^2} = \sqrt{21}$$

$$4\beta^2 + 4 + \alpha^2 + \beta^2 + 2\alpha\beta = 21$$

$$\alpha^2 + 5\beta^2 - 12 = 17$$

$$\alpha^2 + 5\beta^2 = 29 \text{ and } \alpha\beta = -6$$

$$\text{and } \alpha\beta = -6$$

and given α, β are integers

so,

$$\alpha = -3, \beta = 2$$

or

$$\alpha = 3, \beta = -2$$

$$(\alpha_1, \beta_1) = (-3, 2)$$

$$(\alpha_2, \beta_2) = (3, -2)$$

$$\alpha_1^2 + \beta_1^2 - \alpha_2\beta_2 = 9 + 4 + 6 = 19$$